

Making Waves

Recommendations for a Stronger Quantum Ecosystem

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1 Executive Summary

The past 20 years have seen dramatic growth and innovation in software technologies, from the artificial intelligence revolution to the rise of the internet, but over the past few years interest in hardware technologies has spiked. The return of great power competition, the need for technological solutions to climate change and energy security, and the ever-growing need for processing power by software have converged to drive unprecedented levels of investment. Additionally, industrial policy, a strategy thought relegated to developing economies, has been enthusiastically embraced by governments in the developed world. These factors have seen flagship policies like Made in China 2025, the Green Deal Industrial Plan in Europe, and the trio of CHIPS, IRA, and IIJA in the US pouring billions into hardware and manufacturing. Technologies like batteries, renewable energy production, and digital manufacturing are seeing waves of investment. Of these technologies, one of the most important for the future of defense, energy, finance, and more, is one that many have heard only of in science fiction: quantum.

Quantum technologies are at an inflection point. The technology is beginning to exit the research phase, shepherded by industrial policy, but not at the point of being able to stand on its own as a fully-developed, commercial industry. Therefore there is a unique opportunity to shape and accelerate the future of this revolutionary technology, and this report seeks to provide insight into how stakeholders can approach this goal.

This report will begin by diving into the landscape of quantum technologies in one of America's premier hubs, Chicago, and explore recent developments. This discussion will set the stage for the presentation of a series of recommendations that, while applicable to Chicago, are generalizable to other regions seeking to anchor a piece of this world-changing industry. These recommendations are targeted toward three critical pillars that the quantum industry needs to strengthen: talent, focusing on responses to the STEM talent shortage; commercialization, focusing on advancing quantum into an industry with broad applications beyond

traditional governmental ones; and macro stability, seeking to understand what strategies can be employed to protect and support this nascent industry against macroeconomic shocks. The recommendations are as follows:

- **Talent:**

- Universities and companies can expand the supply of quantum talent by integrating private sector on-the-job training into the education model.

- **Commercialization:**

- Non-quantum companies can simultaneously advance quantum technology and accelerate quantum commercialization through open competitions that incentivize the testing, development, and scaling of quantum products with direct impacts on their respective core businesses.
- Governments should work with the financial sector and universities to channel investment dollars into supporting the development and commercialization of quantum products.
- Quantum consortiums can jointly partner to derisk and support commercialization efforts through purchasing quantum-as-a-service products.

- **Macro Stability:**

- Governments should support startup access to funding by counter-cyclically intervening in capital markets and venture ecosystems to shield startups from macroeconomic shocks.

These recommendations are designed to support and grow the quantum industry, oftentimes using proven models in similar industries and uniquely adapting them to quantum. The recommendations also target a wide variety of actors, from various levels of government to universities and private industry, highlighting the importance of broad-based collaboration across all agents in the ecosystem. The application of these recommendations will result in a more robust, technologically and commercially-ready quantum ecosystem, one poised to deliver profound economic benefits and to fundamentally reshape society.

2 Introduction

2.1 Old Physics, New Applications

From silicon transistors to AI, new technologies seem to arise suddenly, ripping through the world and transforming everything in its path. The growth of these technologies in their earlier stages of development, however, was anything but fast. No one could have predicted how the internet would change lives, but, for once, the next technological revolution is clear. How can Chicago position itself to be a global leader in the most revolutionary technology since the computer chip?

Quantum computing, along with communications and sensing, utilizes the principles of quantum mechanics to solve complex problems that ordinary, or classical, computers cannot. Quantum technology is just beginning to emerge as a true area of interest for countries around the world, primarily the United States and China, for its immense computing power. There are already companies claiming complete quantum supremacy over classical computers, and top competitors such as IBM and PsiQuantum expect to have commercialized computers in under a decade. Quantum is no longer a matter of if, but when.

Quantum computers have captured the imaginations of top performers in countless fields for their ability to simulate and optimize. They can probe the subatomic and molecular worlds to create breakthroughs in fundamental physics, protein folding, and drug creation in record time. They can race through optimization problems to allow for financial transactions and global supply chains to run at peak efficiency. Further, their ability to crack the conventional cryptography that currently protects all online information makes quantum an imminent national security risk.

Quantum computing is at a stage where its implementation is no longer an “if” but a “when.” The city or cities that champion this technology early will reap the economic benefits for decades to come. This can come in an influx of high-paying jobs through engineers and programmers but also through more jobs and innovation in manufacturing and investment in that city.

Luckily, Chicago is uniquely suited to be that champion.

2.2 The Capital of Quantum

2.2.1 America’s Quantum Lab

Quantum computing firms certainly require workers with PhDs especially when it comes to high-tech, experimental hardware, but more general needs, such as optimizing algorithms or researching quantum use cases, may not require workers with backgrounds in physics, engineering, or computer science. Chicago is uniquely positioned to grow a quantum workforce through its already established quantum ecosystem, the myriad of top-tier universities in Illinois, and leveraging the idea of Chicago as a premier destination in numerous fields throughout the Midwest.

As of 2021, Illinois is home to four R1 universities – Northwestern University, University of Chicago, University of Chicago Illinois, and University of Illinois Urbana-Champaign – as well as six R2 universities. It is home to four of ten National Quantum Initiative Act centers. Many university partners of the Chicago Quantum Exchange, one of the largest quantum conveners in the nation, reside in Illinois.

Even outside Illinois, Chicago is seen as the destination for many college graduates around the Midwest. Both the University of Wisconsin-Madison and Purdue University in Indiana are a part of the Chicago Quantum Exchange (CQE), and the Midwest region boasts large universities and small colleges throughout. Additionally, the robust partnerships that CQE fosters between companies, startups, and universities are a strong draw for graduates and

researchers from these institutions. Beyond quantum, Chicago’s strength in industries like finance, agriculture, and manufacturing provides ample opportunity for researchers to apply what they’ve developed to a variety of industries and use cases.

2.2.2 Building the Markets of the Future

Chicago has already taken crucial strides to bring quantum to commercialization through partnerships with both industry and government.

The Bloch Tech Hub, led by the Chicago Quantum Exchange, was named a U.S. Regional and Innovation Technology Hub for quantum technologies by the U.S. Department of Commerce in 2023. Thanks to the efforts of the members of the Chicago Quantum Exchange, which include universities and colleges across the Midwest and a variety of startups and nonprofits, bolstered by support from the Illinois state government, Illinois has become a stronghold for efforts to scale quantum computing for both private and government bodies alike. The Defense Advanced Research Projects Agency (DARPA), which is responsible for early investing in revolutionary technologies from GPS to the Internet, has invested \$140 million to build the DARPA-Illinois Quantum Proving Ground (QPG). This investment is matched by another \$140 million from the State of Illinois. The purpose of the QPG is to attract quantum computing companies to Illinois to take advantage of DARPA’s state-of-the-art technology to advance their technologies and prove their ideas can be commercially viable, simultaneously advancing the quantum field through validation that the nascent technology is about to take off. Government-backed spending through the QPG offers quantum a pathway to commercialization while attracting more businesses and spending to Illinois.

Beyond government efforts, Chicago continues to attract private quantum companies. PsiQuantum, a world-renowned quantum startup and a corporate member of the Bloch Tech Hub and the Chicago Quantum Exchange, announced its plans to build a billion-dollar quantum campus in a historically underfunded neighborhood of Chicago. PsiQuantum plans to build the first utility-scale quantum computer in the United States at their new Chicago campus, bringing as many as 150 jobs to the city in the next five years while working with top research universities in Illinois to further strengthen the ecosystems’ research and educational opportunities.

Even outside of its biggest stories, the quantum startup scene in Chicago is particularly strong. Chicago hosts the Duality Accelerator Program, the United States’ first accelerator exclusively focused on quantum innovations. The Chicago Quantum Exchange has also proved vital to fostering connections between labs, industry partners, and incoming talent to bring research dreams to commercial reality. The ecosystem has long been known for its strengths in quantum sensing and communication, and, as shown by the investments made by DARPA and PsiQuantum, its quantum computing scene is growing rapidly and will continue to grow as these campuses further their own developments.

2.2.3 The Jewel of the Midwest

Chicago has a diverse economy with strong financial, agricultural, and manufacturing fields that can reap the benefits of quantum as quantum services become increasingly available. Chicago’s economy is so diverse that no single industry employs more than 14% of its sizable workforce, making it one of the most balanced economies in the United States. Chicago has the third-largest science and engineering workforce of any metropolitan area and the second-largest central business district in the nation. Further, it is ranked as the top metropolitan area for corporate relocation in the United States.

Chicago is a transportation hub in every sense of the word. Chicago is not only one of the most walkable cities in the country, but it also boasts a world-class public transportation system with extensive bus and rail service throughout the metropolitan area. Chicago is also the largest railroad hub in the country and has the world’s second-busiest airport by airline operations.

Lastly, Chicago has a robust and sustainable energy grid. With Illinois being the nation’s largest producer of nuclear energy, and a strong wind presence, energy in Chicago is far more sustainable and green than competitor hubs around the US. Chicago is also a major hub for natural gas, with two natural gas market hubs, and about 10% of national capacity for gas is located in Illinois. This steady supply of gas has been instrumental in decarbonizing the state via the pressure on legacy coal installations and provides reliable, cheap heating during winters. Illinois’ unique combination of nuclear baseline energy, renewables, and gas provides reliable and cheap electricity, with energy per kilowatt-hour in the Chicago metro area costing equal to or less than other peer hubs, and dramatically lower than cities like San Francisco and New York, making it an excellent location for industrial facilities and other energy-intensive industries.

Recommendations

3 Talent

How can we better drive talent into the quantum industry?

3.1 A Leaky Pipeline

Despite the current focus on the subject, labor shortages and questions of talent are an age-old concern. For skilled, technical talent, the crisis is even more acute. For the entire economy, data indicates that demand outstrips supply by nearly 130 thousand positions every year. With the government pouring billions into industrial policy through CHIPS, the Infrastructure Investment and Jobs Act, and the Inflation Reduction Act, demand and competition for STEM talent will only intensify.

This problem is even worse for nascent fields like quantum, where smaller shortages in absolute terms can have a devastating impact on the relatively small field. A survey from

the Quantum Economic Development Consortium (QED-C) found that 92% of companies said there was a shortage of domestic quantum talent, and nearly 60% reported that they failed to meet their hiring targets. Furthermore, the smaller size and less developed nature of the industry means that many firms cannot command the financial resources needed to compete with larger employers' demand for technical talent, with much of the best talent choosing high-paying fields like tech. Even worse, less than 40% of graduated engineers pursue a career in engineering, indicating leakage towards other high-paying sectors like finance. These shortages stifle research progress and introduce costly delays in the process.

The problem persists not for lack of trying, however. However, little attention has been paid to better capturing and securing the existing engineering pipeline. With around 200 thousand engineers graduating per year, reducing the attrition rate could result in dramatically more engineers in technical fields. If the quantum industry could capture these gains, even a 1% reduction in the attrition rate would mean nearly 2000 additional engineers per year entering the field, a dramatic influx of talent. Given the sheer size of the problem, what can the industry do to begin to tackle this challenge?

3.2 Recommendation 1

Universities and companies can expand the supply of quantum talent by integrating private sector on-the-job training into the education model.

One model that companies could implement would be to offer fellowships for promising university talent. These programs would pay for up to four years of undergraduate tuition and provide the students with paid internship opportunities throughout their summers. They would then require the student to work at the funding company after graduation, for example, for 1.5 years per year of funding. Companies could offer these fellowships in disciplines like computer science/engineering, materials engineering, physics, and so on, aligning with whatever talent needs they have, providing companies with a steady stream of university talent by using a proven model (DoD's SMART program).

It should be noted that this recommendation only covers part of the quantum workforce, namely engineers. When it comes to those responsible for connecting quantum technologies to non-quantum businesses, business liaisons who don't have a deep background in science are necessary for building connections outside the ecosystem. Also, there are applications for non-quantum talent to explore quantum use cases throughout various industries and research fields. Finally, there is a variety of talent and labor needed in construction and technicians, and anyone looking to truly bolster every aspect of the quantum workforce needs to keep these jobs in mind.

3.2.1 Broaden the Base

Four-year universities are not the only source of talent for engineers, and opening up more sources would go a long way in filling these labor shortages. Community colleges already

have a ‘two and two’ model, where students study for two years at a community college and then transfer to a four-year institution to finish their degree. An expansion of this model could involve both private and college action: community colleges could, in conjunction with interested private parties, actively seek and recruit students, helping them receive a private fellowship upon transfer to a four-year institution. This would result in not only higher quality talent as the best of the best from every source are considered, but also prevent the gains from being concentrated in any one community.

This model also broadens the communities from which the talent is sourced. Evidence indicates that debt from college pushes students (especially lower-income students) into initially higher-paying but lower overall lifetime earning positions to pay for the debt in the short term. Quantum’s position as a nascent field poised to make significant future breakthroughs makes it a prime example of an industry with high-income growth but potentially lower starting earnings. Additionally, by actively recruiting from those schools, awareness is created which would have positive spillover benefits in encouraging more community college students to consider careers in quantum.

3.2.2 Lower the Attrition Rates

By providing fellowships to incoming students or students mid-way through their education, the funding company gets an educated engineer for a guaranteed length of time, which will begin to solve one of the primary problems in the talent shortage: attrition of engineers. The paid opportunities over the summer also allow for more specific training, reducing or eliminating the need for on-the-job training or for the student (now employee) to become accustomed to working at the firm.

Additionally, the job security provided by the program combats a fear students face when considering entering quantum: uncertainty. Anchoring these students in the nascent field for an extended period (before and after graduation) provides a basis to build skills and experience without fear of job loss. Even in the circumstance they do lose their job, the employee will have had enough time to build a resume that can get them hired elsewhere. Lastly, this security solves an issue that arises when trying to encourage low-income students to participate in these models: the problem of immediate funding. If a student cannot afford to make rent while waiting for a job down the line, a much higher risk of attrition is likely. By providing guaranteed paid internships over the summer and sourcing jobs for students before graduation that activate immediately when the student is ready, the amount of time where this could occur is drastically minimized, helping keep more diverse talent in the model and bringing down the attrition rate.

3.2.3 Positive-sum Investments

The primary barrier to the implementation of this model would be the perceived cost. However, a rough calculation ¹ of this model’s four-year net present cost per student is approxi-

¹Calculated using nominal dollars at an interest rate of 3.94%, the 10-year Treasury rate as of writing. Costs include four years of tuition at the average cost of the respective school type and three summers of internships. Internships are assessed at 10 weeks of full-time work at \$15 per hour. Community colleges are

mately \$165K for private non-profit schools and \$52K for public schools, including the cost of paid summer opportunities. For a firm’s investment to break even a private school student would need to generate about \$67K of additional value per year for their six years and a public school student \$22K per year, achievable values for staff at startups with capital raises in the hundreds of thousands or millions or large companies with R&D budgets. These costs would be even lower for community college students, at \$10K a year of additional value for three years, creating a further incentive to recruit from these institutions. For large and small companies, hiring students at this cost would be an efficient way to attract talent, and leveraging existing relationships with universities may allow them to get a better deal. Even in the absence of a discount this model would likely be cheaper than attempting to poach an experienced professional from a more salary-competitive industry or having to give out equity to do so.

Lastly, there are strong benefits for regions, universities, and community colleges, and this may make implementation easier as these entities cooperate. Regional ecosystems would benefit by drawing or retaining technical talent in their quantum ecosystem and would see agglomeration effects from the added talent working together in a shared ecosystem. Universities and community colleges benefit by being able to point to such programs as a selling point to prospective students. Community colleges specifically would see greater awareness of these careers in the communities they serve, helping drive further enrollment into these programs. Every party benefits from developing a reputation as a top retainer of quantum talent.

Quantum is a rapidly changing industry, thus the amount of talent and the skills they are taught need to be sufficient to adapt and pivot as needed. By integrating employers into the education and training model and enabling new talent to join the workforce, the industry can begin to align labor supply with demand and help ensure the students of today, from any background, make the discoveries of tomorrow.

4 Commercialization

How can we begin to move quantum products into a commercial-ready state?

4.1 An Uncertain Value Proposition

The science behind quantum technology has been understood for decades, but only recently has that knowledge begun to be applied to usable commercial products. The US government has funded this basic research at the university and national lab level for years and recently has started pouring money in through the passage of acts like the National Quantum Initiative and CHIPS. Despite the progress being made in the fundamentals of the technology, much of industry outside of quantum remains skeptical of the potential use cases. With quantum currently requiring high initial set-up costs for unknown gains, industry leaders are by and large refraining from making large-scale investments in this technology. This

assessed at two years of public tuition and one summer of internship.

is a problem for the advancement of the field because until robust commercial demand exists for quantum products, the industry will be reliant on early-stage industrial policy and defense demand, relegating the industry to a niche position. To create and drive demand for quantum products, industry needs to think about applications of quantum in their core business.

One problem is that many industry leaders may not know where to begin. Quantum is a highly complicated field, quantum experts are also few and far between, and companies are unlikely to retain talent simply to understand what a fledgling technology could theoretically do in the future. These issues make implementation difficult since industry leaders don't know what is feasibly worth investing in and where to focus attention.

Further, venture funding is critical to commercialization and serves as a multiplier to the efficacy of government and other R&D spending. A greater supply of venture funds is shown to increase patenting and firm creation, but a lack of venture funding can result in technology being stuck in its infancy or startups struggling to expand.

These issues all present critical threats to the advancement of the ecosystem and society as a whole, as the vast majority of the benefits from quantum are locked behind commercialization and rapid scaling of the technology. If industries intend to harness the value of this vital technology, how can they work with the quantum ecosystem to develop products that serve their needs without the need for massive upfront in an uncertain technology?

4.2 Recommendation 2

Non-quantum companies can simultaneously advance quantum technology and accelerate quantum commercialization through open competitions that incentivize the testing, development, and scaling of quantum products with direct impacts on their respective core businesses.

A powerful tool that companies have to achieve this goal is advanced market commitments (AMCs). An AMC is a promise to buy a product down the line, provided it meets certain criteria, essentially a competition where the prize is a guaranteed sell option to the company issuing the AMC. The power of this model was famously demonstrated by the prediction-defying development of the COVID-19 vaccine through Operation Warp Speed.

Companies can harness the power of AMCs for quantum by offering to buy certain quantum products at a future point in time. These commitments would specify a benchmark or target improvement for the product to hit, and the first person or company to create a product at or above that benchmark would have a guaranteed sell option to the company issuing the AMC. The guaranteed sell option incentivizes anyone who is working in that space to work towards a solution tailored to the company's need since it's guaranteed revenue.

4.2.1 Advancing the Marketplace of Ideas

There are a few powerful benefits to this model over traditional R&D funding. The first is that it’s actor-agnostic. Anyone is capable of cashing in on the AMC, whether it’s a startup, a large company, or even a single inventor. This diversity of R&D oriented toward solving the problem is more effective than funding a single entity. This is especially helpful in the context of quantum, with the field consisting of everything from legacy technology companies like IBM to startups spun out of graduate programs.

Another benefit is that it’s also method-agnostic. The mechanics behind the product are irrelevant, only whether it meets the specified criteria. An example of this would be for quantum computers. A company could want to buy a future quantum computer capable of computing a process in a certain amount of time. Traditional R&D spending would require the company to pick a specific type of quantum computer. An AMC means that any technology could win out, letting a wider variety of potential technologies be spun into a commercial product and avoiding ‘picking winners and losers’, a common concern of industrial policy. By keeping the technology and firm pathways open, AMCs don’t distort the research process and let the best technology and firm win, preventing path dependency or locking into a sub-optimal research path.

While government-led AMCs have been recommended for quantum, the corporate-led AMCs make this recommendation distinct from previous ones. Government AMCs typically focus on a product with governmental applications, such as defense or public health. A corporate AMC is unique in that it drives a product that the private sector can use. This way, it not only pulls the technology forward but also integrates it with industry, creating the basis for further demand for quantum products as competitors see the benefits gained. This also means that this idea is more powerful in hubs with a diversity of industries, since it allows for a wider variety of AMCs to be issued, pulling forward different technologies. For example, manufacturing and agriculture could issue AMCs for very precise quantum sensors to detect field conditions or defects, helping advance that aspect of quantum as well.

4.2.2 Derisking R&D

Additionally, an AMC is a near-zero-financial risk option for companies beyond the time and capacity to develop the AMC. An AMC only pays out once the product is verified to meet the company’s requirements, and a profit-maximizing company will only select benchmarks and purchase prices it projects will be profitable. On the flip side, if no product is developed that meets the company’s standards, no expense is incurred since no one is eligible to claim the AMC.

AMCs are also a good deal for the inventing companies. If they reach a product that meets the specifications, there is a guaranteed payout which provides certainty for investors and the company. This also would likely allow companies to raise funds based on the promise of meeting a future AMC. Even if the company does not reach the AMC, there may be innovations made in the process of trying to get there that can be commercialized.

The diverse nature of the quantum industry is a strong asset in finding and developing the

best possible technologies and R&D strategies should take advantage of that. By implementing a model that pulls the technology forward, lets the best idea come to the forefront, and commercializes it, industry can begin to reap the benefits of this extraordinarily valuable technology.

4.3 Recommendation 3

Governments should work with the financial sector and universities to support and channel investment dollars into commercializing quantum products.

The specifics of implementation can and should vary by region, but a general approach that regions can apply is modeled after the Biomedical Advanced Research and Development Authority Division of Research, Innovation and Ventures' (BARDA DRIVE) approach in the life sciences realm. A state or federal agency could create a joint public-private pool of funds, with a corresponding quantum mission space. The agency would designate a private venture capital partner to lead the investment, with public funds matching other private funds raised.

This approach is designed to derisk venture capital investment into admittedly risky technologies, without needing to appropriate a large sum of money every year. The capital fund is also a one-time investment and is designed to grow on its own as the industry grows, without additional appropriations. By having the government underwrite some of the risks, this model is proven in other realms to successfully drive investment into and commercialize technologies. It also has the benefit of being private-led, avoiding some concerns about market distortion and other typical industrial policy concerns. It also allows for venture firms to get experience in investing in this market, potentially crowding in more venture investment, akin to how federal R&D dollars drive further private investment.

4.3.1 A Modular Approach

Beyond the joint pool, there are various additional strategies that regions could pair with this model to expand its impact. The first is providing additional technical and regulatory support to startups in the joint portfolio. These could be in-house talent or contractors but would work with the firms to resolve technical challenges and advance the firm towards the goals laid out by the agency and investors. Additionally, having staff on hand who are knowledgeable about how existing regulations apply and where the landscape might evolve would be greatly beneficial in providing regulatory certainty for founders.

This joint pool could also attract out-of-state or out-of-region venture firms. This would be of greatest help to regions without a large venture ecosystem, but would still be beneficial anywhere. There would be no worry that any additional investment into quantum comes at the expense of investment into another industry in the region, for there would be entirely new dollars that add to the ecosystem. The entry of the firm itself also has significant benefits for the regional venture ecosystem. More venture capital firms mean more competition for

deal flow, creating more favorable terms for founders, and lowering the hurdles needed to get a startup off the ground.

One of the most important benefits of this addition is the potential for knowledge spillover. Diverse venture capital portfolios increase firm innovation and the chance for cross-disciplinary breakthroughs, which are responsible for many of the most significant advancements in recent history. For example, the combination of hardware advances in the form of GPUs enabled neural networks to be feasibly run and improved, beginning the massive improvement seen in that field. By drawing non-regional venture firms into the ecosystem, quantum firms can then benefit by being able to access ideas and innovations from the diverse industries in the portfolios of the new venture entries.

4.3.2 The Yale Model of Commercialization

The ‘Yale Model’ in investing refers to the strategy developed by the managers of the Yale University endowment to invest those dollars into private markets like private equity and venture capital. This strategy has been widely adopted by many universities, but especially large private and/or research universities. Thus, there exists an opportunity to align endowments with the joint pool to further commercialization goals.

By offering universities the option to participate in the joint pool, many of the same benefits that apply to a conventional venture fund would also apply here, drawing an endowment manager to make such an investment. For example, since many endowments are more risk-averse, the government’s involvement in the fund may serve to assuage concerns and serve as an additional reason to join the investment pool. Also, funds that have more human capital managing it tend to perform better, and specifically improve the performance of investment in alternative assets like venture. There are also significant branding benefits, as involvement in the fund and a stronger regional venture ecosystem are strong attractors to potential students, donors, and government R&D.

Venture capital is not a silver bullet, and simply channeling funds into quantum will not by itself convert the industry into a commercial-ready one. But by connecting experienced talent with operators, and funding with startups, agents can begin to construct a well-formed ecosystem that provides the basis for a self-sustaining, commercial-ready quantum industry that can eventually stand without the support of industrial policy.

4.4 Recommendation 4

Quantum consortiums should support commercialization efforts by agreeing to jointly purchase quantum as a service products.

One approach is for state university systems, private universities, or other members of quantum consortiums to jointly purchase quantum as a service (QaaS) products from a single provider. These groups can require the QaaS provider to relocate to the region in which the

consortium resides or, depending on the need and request of the consortium, build a facility or open an office in that region.

QaaS companies have access to quantum computers but, instead of providing physical quantum technology, it provides a specific quantum service at the buyer's request. This approach is designed to allow users to try quantum products, and begin to think about what needs they have that can be solved by these new technologies, without substantial upfront investment.

4.4.1 Quantum Wholesale

To scale QaaS, a guaranteed customer base is needed, one that the joint purchase will guarantee. In cities that are already quantum technology hubs, there are likely both companies and institutions, such as universities or labs, that can greatly benefit from access to a quantum computer and are also connected to a broader ecosystem that supports investment in QaaS. This model also allows companies to think about where future demand can come from by jump-starting the regional adoption process of quantum technologies.

QaaS also provides a collaborative process between the provider and buyer as they work to understand the buyer's needs through a quantum lens. As the provider meets more and more of these needs, the buyer will gain a greater understanding of the abilities and strengths of quantum computing and will continue to buy from the QaaS provider. This long-lasting need guarantees a customer base, and as the institutions reap the benefits of quantum computing, others may join in buying QaaS products so as not to be left behind. Also, because much of that initial customer base is likely to be universities and research institutions, it will benefit the QaaS company to be close to where a lot of skilled talent is. And, as these universities may have already introduced their students to quantum computing use cases, students may be more likely to want to work in a quantum field. Lastly, the regional agglomeration effects cannot be ignored. Developing a need for regional colocation of quantum products will concentrate research and jobs in quantum, thereby putting researchers together and taking advantage of scale effects to see increased growth in the ecosystem.

QaaS is an important part of introducing quantum-skeptical users to what this technology can do for them, a key step in beginning the broader adoption of quantum products overall. By utilizing the power of group purchasing, regions can use QaaS companies as the seed for a much broader quantum industry.

5 Macro Stability

How can we protect the quantum industry from macroeconomic shocks?

5.1 The Ultimate Arbiter

Quantum is still nascent and being guided by industrial policy and government intervention. The traditional path is for the industry to advance beyond these supports and to achieve a self-sustaining commercial pipeline of business to become a full-fledged industry. While this

approach has been successful in the past, there have also been notable failures, largely due to macroeconomic factors.

A prime example of this is batteries. Lithium Iron Phosphate (LFP) batteries occupied a similar position, with governments investing in industrial policy intended to guide the growth of this industry. With the primary application being for electric vehicles, it's easy to see why this technology is so important. The support from the government resulted in numerous startups focusing on LFP technology, with some achieving spectacular success. These startups founded multiple domestic and international factories, had various commercial deals with large established companies, and were on track to become the lynchpins of a new American battery industry. But despite these successes, today China dominates in this technology. A worsening economic situation after the Great Recession crushed the prospects of these startups and only recently has the US started to rebuild this industry. Quantum is a more nascent industry than batteries were, and the race between the US and China for this key technology is even tighter. A localized shock in the US could severely damage the ecosystem, ending hopes of chances for winning the quantum race.

Beyond batteries, there is evidence that recessions in the real economy and reduced capital flows result in less early-stage investment, where most quantum companies are. In the case of recessions, investors move capital into safer assets, and reduced capital flows signal negatively about the performance of IPOs, causing investors to prefer to refinance later-stage startups. Quantum's newness would compound the effect of flight from relatively riskier investments, resulting in a shift out of quantum entirely into safer assets. These risks are especially worrisome because they could result in human capital hysteresis, where managers experienced in quantum lose skills when shifted into other industries, and knowledge within quantum is lost as deal flow dries up.

Beyond funding constraints, macroeconomic downturns can also result in various problems. During downturns, higher-quality talent flees to more established firms, starving startups of staff. Economic downturns can also cause more established companies to cut R&D spending on nascent technologies or for potential customers of startups to pull back on making riskier investments or purchases.

Even macroeconomic booms can result in issues for startups. The boom in economic activity following the COVID-19 pandemic saw a rapid rise in interest rates to combat inflation, and despite the strong overall economic performance of the past few years, venture capital funding and equity fundraising deals have collapsed as a result of the increased rate pressure. As the hurdle rate rises, more and more startups will fail to clear this, constricting the ecosystem.

Quantum startups already face numerous challenges in trying to successfully commercialize and achieve profitability. History shows that even with traditional industrial policy measures, more work is needed to explicitly counter the effects of broader macroeconomic barriers. Given that both good times and bad times can have negative effects on the quantum ecosystem, how can governments construct an approach that is flexible enough to handle both cases?

5.2 Recommendation 5

Governments should support startup access to funding by counter-cyclically intervening in capital markets and venture ecosystems to shield startups from macroeconomic shocks.

This approach is modeled after DOE’s approach to try and keep oil prices stable. DOE used the Strategic Petroleum Reserve (SPR) to release oil onto the market when prices were high, helping bring down prices and issuing a buy order at a lower price to create a price floor, providing certainty for oil producers. In a similar model, agencies should set a target investment flow benchmark for quantum technologies, and should funding approach or fall beneath the benchmark, the government would make up the shortfall, either by itself or using the previously mentioned joint ventures model. Once investment flows rise and return to normal, the fund could decrease investment and begin to liquidate its positions. Governments can support this by using contracting authority and interagency investment to provide contracts and additional funds.

5.2.1 Moderating Markets

This approach has various benefits. For early-stage firms, government intervention would make up some of the shortfall from the loss of venture funds and help provide some of the beneficial network effects to startup growth. It would also provide a hedge against macroeconomic conditions. Firms would be less likely to take extreme measures in anticipation of a crisis or to accept unfavorable terms from a lender because the intervention keeps capital markets flowing as usual. For later-stage startups, coalition investment from governments can keep the contract pipeline flowing. Quantum is a great industry for this since it has direct intelligence and defense applications, so DoD or the intelligence community could support counter-cyclical efforts by devoting more budget to startup contracting.

This approach is also applicable to a rate shock situation, as the government can offer deals slightly below the market rate. This injection of capital can maintain otherwise robust startups that currently struggle to raise capital in business long enough for interest rates to come back down. This offering can also bid down the rates in the market or ecosystem by reversing some of the dynamics of a credit crunch.

5.2.2 Real Benefits

Another benefit for startups is the impact it would have on talent. Startups often lose out on the best talent when the economy worsens, but with sufficient capital on hand, startups may be able to compete for this talent. In a poor labor market, there would be a significant decrease in hiring, and additional funds may allow startups to offer compensation closer to what the labor market as a whole offers.

More broadly, recessions create opportunities to support higher-quality startups. Since decreased salaries lower the opportunity cost of starting a company, many top employees will choose to instead become entrepreneurs and founders. By surging capital, it becomes easier

to support new firms and possibly speed up the success of the startup by reducing financing constraints.

Lastly, there are significant benefits to the government. Beyond the effects of supporting a strong quantum industry and the material impacts of the technologies developed, the government can generate revenue and/or a self-sustaining fund. By investing when the market isn't, the government can find hidden opportunities that would be otherwise missed. Then when the market picks up, the government can sell these positions for a profit just as DOE's 'buy low, sell high' approach for oil works. This approach also supports broader macroeconomic stabilization goals, as increased investment during a recession helps the economy recover faster and decreased spending during booms moderates inflationary pressures.

Access to finance is indisputably among the most important determining factors of the success of a startup, but oftentimes that access is unmoored from the actual prospects of the firm. By counter-cyclically investing in capital markets and the venture ecosystem, the government can balance these at extreme moments, supporting good deal flow, encouraging and supporting startups, and ultimately accelerating the progress of the quantum industry.

6 Additional Considerations

6.1 All-Purpose Power

A crucial aspect of the quantum landscape that has not been mentioned in this paper previously is the cybersecurity aspect. An error-corrected quantum computer will have the capability to break current models of standard encryption. Given that the United States is not the only country in the race for quantum supremacy – China, in particular, has advanced quantum technologies and has devoted a large amount of government funding to quantum – the advent of a foreign quantum computer, or even the access to a quantum computer by an ill-intentioned party, is a national security risk. Even encrypted data today is not safe from future quantum computers. Many hackers have a “mine first, decrypt later” mentality to acquiring data, so information with a long lifespan, for example, medical data, is not secure from a quantum threat in the long run. This is a major security risk that government and industry alike need to be actively thinking about.

The good news is that protecting data from quantum computers does not actually require a quantum computer. There are current post-quantum cryptography (PQC) standards that have been released by the National Institute of Standards and Technology (NIST). Because of the existential risk quantum computers pose to all data, PQC algorithms ought to be implemented in every company as early as possible, but this is easier said than done. Switching from standard cryptography to PQC requires companies to complete an expensive overhaul of a system that requires greater computing power. Though NIST has current recommendations, there are likely more updates and fine-tuning to follow, potentially disrupting security transitioning attempts further. Though PQC is hard to implement, the risks of not switching to it need to be well known.

One final risk to consider is the direct security of quantum assets, especially given the fractured and mostly commercial development of these technologies. Already many companies developing other technologies like AI and semiconductors have seen foreign adversaries attempt to steal IP through economic, personnel, and more direct means. As quantum becomes an increasingly integral part of great power competition, developing strategies to coordinate governmental and industry actors to safeguard these crucial technologies without needlessly impeding the industry from innovation will be an important task, one that will determine whether quantum's benefits are deployed in support of friend or foe.

6.2 Supply-side Security

Another piece of a functioning quantum ecosystem that is not often talked about is supply chains. This is not a pressing issue yet because of how small the quantum arena is right now, but because of the growth quantum technologies are expected to undergo, the lack of a quantum supply chain will become an increasingly glaring issue. When quantum computers are commercialized, defense-ready supply chains need to be in place. A global quantum supply chain should not have the United States relying on competitor nations that could steal IP. Instead, factories should be set up in friendly countries or domestically, where IP is safe and cooperation will benefit all involved economies. Governments and industries should begin thinking about how to set up factories as soon as possible and, in the case of foreign factories, how to protect IP while ensuring cooperation between involved nations.

6.3 Conclusion

It cannot be emphasized enough how exciting the quantum space is right now. The people with active roles in their ecosystems are shaping a future multi-billion-dollar industry, which is an incredibly unique position to be in. There is so much passion and excitement in every individual in an arena that seems unimaginably big on a national and international scale. But this is only the first step - the implementation of these recommendations, and many other challenges that the ecosystem needs to address will require a major systems change, culture shift, or overhaul as a prerequisite. The private sector, from entire industries to individual companies, need to be open to quantum solutions. The future of quantum will be shaped by how different actors cooperate to drive the technology forward. And, above all, there needs to be those individuals spreading the word, who recognize the awesome potential as well as the possible pitfalls of quantum, and who can't stop talking about it.

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